LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Most-Expected Questions: Chemistry (Ch 1–4)

Chapter / Topic: Chemical Reactions · Acids, Bases and Salts · Metals and Non-metals · Carbon and its Compounds

Name: _____ Roll No.: _____ Date: _____

Marks obtained: _____

General instructions:

1. These are the question types that come up again and again in board papers. Answer them in your notebook, then check with the answer key.
2. Tick each question when you can answer it without looking. Come back to the unticked ones a week later.

1-MARK QUESTIONS

Q1. What do you observe when barium chloride solution is added to sodium sulphate solution?

Q2. Name the type of reaction: $CaCO_3 \rightarrow CaO + CO_2$ (on heating).

Q3. What is the common name of $CaOCl_2$?

Q4. What happens to a turmeric stain on a cloth when soap is applied?

Q5. Name two metals that do not react with oxygen even at high temperatures.

Q6. Why is graphite used to make electrodes?

Q7. Name the functional group in CH_3COCH_3.

Q8. Name two metals that are found in the free state in nature.

2-MARK QUESTIONS

Q9. Potassium chloride solution is mixed with silver nitrate solution, and an insoluble white substance forms. Write the balanced equation and name the type of reaction.

Q10. Five solutions A, B, C, D and E have pH 4, 1, 11, 7 and 9. Which is (a) neutral, (b) strongly alkaline, (c) strongly acidic, (d) weakly acidic, (e) weakly alkaline? Arrange them in increasing order of H^+ ion concentration.

Q11. Why do HCl and HNO_3 show acidic character in water, while solutions of alcohol and glucose do not?

Q12. What are endothermic and exothermic reactions? Give one example of each.

Q13. Why is sodium stored under kerosene, while white phosphorus is stored under water?

Q14. Describe the bonding in (a) carbon dioxide and (b) nitrogen.

Q15. What is formed when a metal oxide reacts with an acid? Give an example with a balanced equation.

Q16. Explain the bonding in CH_3Cl (chloromethane).

3-MARK QUESTIONS

Q17. A compound made from gypsum hardens when mixed with the right amount of water. Identify it, write the equation for its preparation, and give one use.

Q18. Give reasons: (i) Ionic compounds have high melting points. (ii) Calcium floats when it reacts with water. (iii) Copper sulphate solution cannot be stored in an iron vessel.

Q19. An element reacts with oxygen to form a compound with a high melting point that is also soluble in water. What could the element be? Explain.

Q20. What is a homologous series? Explain using the first three members of the alcohol family.

Q21. Write the balanced equations for what happens when (i) zinc carbonate is heated strongly, (ii) zinc oxide is heated with carbon, (iii) zinc is added to copper sulphate solution.

Q22. Why is it said that carbon forms a very large number of compounds? Give three reasons.

5-MARK QUESTIONS

Q23. (a) Arrange Mg, Na, Zn, Cu and Fe in decreasing order of reactivity. [1]
(b) An iron nail is put in zinc sulphate solution and another in copper sulphate solution. In which will a reaction take place? Write the equation. [2]
(c) Why are gold and silver used for jewellery? [2]

Q24. (a) Write the equations for the preparation of bleaching powder, baking soda and washing soda. [3]
(b) Give one use of each. [2]

Q25. (a) What is saponification? Write the equation for ethyl ethanoate with sodium hydroxide. [2]
(b) How do soaps clean? [2]
(c) Why do detergents work in hard water? [1]

~ If you can answer all of these without the book, you're board-ready for Chemistry! ~